

MMM Basics Part 1

Points: 23/50 Time: 19:59

1. Email *

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2. Name *

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✗ **Incorrect** 0/1 Points

3. **Bayesian MMM uses Markov Chain Monte Carlo (MCMC) for:**

- ☐ a) Sampling posterior distributions
- ☒ b) Calculating priors
- ☐ c) Reducing computation time
- ☐ d) Validating hypotheses

✓ **Correct** 1/1 Points

4. **What is the “base” in base + incremental sales decomposition?**

- ☐ a) Sales from marketing
- ☐ b) Sales from promotions
- ☒ c) Sales that would have happened without marketing
- ☐ d) Sales from new products

✓ **Correct** 1/1 Points

5. **What does “saturation” mean in the context of MMM?**

- ☐ a) When all customers have been reached
- ☒ b) When additional spend yields diminishing returns
- ☐ c) When sales are at their peak
- ☐ d) When media reach is 100%

✗ **Incorrect** 0/1 Points

6. **Which of the following is a typical data source for MMM?**

- ☒ a) Social media posts
- ☐ b) POS (Point of Sale) data
- ☐ c) Employee satisfaction surveys

☐ d) Focus group transcripts

✗ **Incorrect** 0/1 Points

7. **What is the purpose of a “holdout sample” in MMM validation?**

☐ a) To test model performance on unseen data

☐ b) To increase the sample size

☐ c) To reduce overfitting

☐ d) To create more predictors

✓ **Correct** 1/1 Points

8. **Which of the following is a common output of MMM?**

☒ a) Contribution curves

☐ b) Customer segments

☐ c) Brand equity scores

☐ d) Creative recommendations

✓ **Correct** 1/1 Points

9. 1. **Which of the following is a key challenge when using MMM for digital channels?**

a) Lack of data

b) Shorter lag effects

- c) Attribution overlap with other channels
- d) No diminishing returns

- ☐ a) Lack of data
- ☐ b) Shorter lag effects
- ☒ c) Attribution overlap with other channels
- ☐ d) No diminishing returns

✓ **Correct** 1/1 Points

10. Question **Which of the following is a reason to use Bayesian methods in MMM?**

- ☒ a) To incorporate prior knowledge
- ☐ b) To speed up computation
- ☐ c) To avoid using control variables
- ☐ d) To remove all uncertainty

✓ **Correct** 1/1 Points

11. **What is the main difference between MMM and Multi-Touch Attribution (MTA)?**

- ☐ a) MMM uses person-level data, MTA uses aggregate data

- ☒ b) MMM uses aggregate data, MTA uses person-level data
- ☐ c) MMM is real-time, MTA is not
- ☐ d) MMM ignores offline channels

✓ **Correct** 1/1 Points

12. **What is the main benefit of using regularization techniques in MMM?**

- ☐ a) To handle missing data
- ☒ b) To reduce overfitting
- ☐ c) To speed up computation
- ☐ d) To increase the number of predictors

✓ **Correct** 1/1 Points

13. **Which of the following is a typical way to visualize MMM results?**

- ☒ a) Response curves
- ☐ b) Pie charts
- ☐ c) Scatter plots
- ☐ d) Heat maps

✓ **Correct** 1/1 Points

14. **Which of the following is a common approach to handle seasonality in MMM?**

- ☒ a) Add seasonal dummy variables
- ☐ b) Ignore it
- ☐ c) Use only non-seasonal data
- ☐ d) Remove all holidays

✓ **Correct** 1/1 Points

15. **Which of the following is a potential bias in MMM?**

- ☐ a) Omitted variable bias
- ☐ b) Sampling bias
- ☐ c) Measurement error
- ☒ d) All of the above

✓ **Correct** 1/1 Points

16. **What is the purpose of “lagged variables” in MMM?**

- ☒ a) To model the delayed effect of marketing
- ☐ b) To model seasonality
- ☐ c) To model base sales
- ☐ d) To model competitor effects

✓ **Correct** 1/1 Points

17. **What is the main reason to use a logarithmic transformation in MMM?**

- ☒ a) To linearize non-linear relationships
- ☐ b) To reduce the number of variables
- ☐ c) To increase model complexity
- ☐ d) To remove seasonality

✗ **Incorrect** 0/1 Points

18. **Which of the following is a limitation of MMM in a digital-first world?**

- ☒ a) Cannot measure offline channels
- ☐ b) Attribution overlap with digital channels
- ☐ c) Too much granularity

☐ d) No need for control variables

✓ **Correct** 1/1 Points

19. **Which of the following is a key limitation of using linear regression for MMM?**

☐ a) Inability to handle non-linear relationships

☐ b) Overfitting due to high-dimensional data

☐ c) Sensitivity to outliers

☒ d) All of the above

✗ **Incorrect** 0/1 Points

20. **. Which transformation is most appropriate for a zero-inflated media variable?**

☐ a) Log transformation

☐ b) Square root transformation

☐ c) Min-Max scaling

☒ d) Adstock transformation

✗ **Incorrect** 0/1 Points

21. **Which validation technique is most robust for time-series MMM data?**

- ☐ a) k-Fold cross-validation
- ☐ b) Holdout sample validation
- ☐ c) Walk-forward validation
- ☐ d) Leave-one-out validation

✗ **Incorrect** 0/1 Points

22. **.A media channel with a negative ROI should:**

- ☐ a) Be excluded from the model
- ☐ b) Have its budget reallocated
- ☐ c) Be retested with interaction terms
- ☐ d) All of the above

✓ **Correct** 1/1 Points

23. **The Durbin-Watson statistic tests for:**

- ☐ a) Multicollinearity

- ☒ b) Autocorrelation
- ☐ c) Heteroscedasticity
- ☐ d) Normality of residuals

✗ **Incorrect** 0/1 Points

24. **Which machine learning method is least suitable for MMM?**

- ☐ a) Random Forest
- ☐ b) LASSO regression
- ☐ c) Gradient Boosting
- ☐ d) k-Nearest Neighbors

✗ **Incorrect** 0/1 Points

25. **The primary purpose of a "data clean room" in MMM is to:**

- ☐ a) Remove outliers
- ☐ b) Anonymize customer data
- ☐ c) Enable privacy-compliant data collaboration
- ☐ d) Normalize variables

✗ **Incorrect** 0/1 Points

26. **Which external factor is least likely to be included in MMM?**

- ☐ a) GDP growth
- ☐ b) Competitor pricing
- ☐ c) Social media sentiment
- ☐ d) Employee turnover rate

✗ **Incorrect** 0/1 Points

27. **A Bayesian MMM using MCMC (Markov Chain Monte Carlo) aims to:**

- ☐ a) Reduce computational time
- ☐ b) Sample from posterior distributions
- ☐ c) Eliminate prior distributions
- ☐ d) Optimize media spend

✓ **Correct** 1/1 Points

28. **Which metric is most critical for evaluating MMM forecast accuracy?**

- ☐ a) R-squared
- ☒ b) MAPE (Mean Absolute Percentage Error)
- ☐ c) F-statistic
- ☐ d) p-value

✓ **Correct** 1/1 Points

29. **The "halo effect" in MMM refers to:**

- ☐ a) A channel's impact on non-target metrics
- ☐ b) Overestimation of ROI due to biased data
- ☒ c) Brand equity spillover to related products
- ☐ d) Seasonal sales peaks

✗ **Incorrect** 0/1 Points

30. **Which approach best handles missing data in MMM?**

- ☐ a) Listwise deletion
- ☐ b) Multiple imputation
- ☐ c) Mean substitution
- ☐ d) Forward-filling

✗ **Incorrect** 0/1 Points

31. **A coefficient of 0.3 for TV spend in a log-log model implies:**

- ☐ a) 10% increase in TV spend → 3% sales lift
- ☒ b) 1% increase in TV spend → 0.3% sales lift
- ☐ c) 0.3% sales lift per TV GRP
- ☐ d) Non-linear relationship

✓ **Correct** 1/1 Points

32. **The primary advantage of using ridge regression in MMM is:**

- ☐ a) Handling non-linearity
- ☒ b) Reducing overfitting via L2 regularization
- ☐ c) Feature selection
- ☐ d) Modeling interactions

✗ **Incorrect** 0/1 Points

33. **. In MMM, "concurrent effects" refer to:**

- ☐ a) Immediate sales impact of marketing
- ☐ b) Lagged effects from prior periods
- ☐ c) Synergy between online and offline channels
- ☐ d) Seasonal trends

✓ **Correct** 1/1 Points

34. **Which diagnostic plot is most useful for detecting heteroscedasticity?**

- ☐ a) Q-Q plot
- ☒ b) Residuals vs. fitted values
- ☐ c) Histogram of residuals
- ☐ d) ACF plot

✗ **Incorrect** 0/1 Points

35. **The Dickey-Fuller test checks for:**

- ☐ a) Stationarity in time-series data
- ☒ b) Normality of residuals
- ☐ c) Multicollinearity

☐ d) Outliers

✗ **Incorrect** 0/1 Points

36. . A log-likelihood ratio test is used to:

- ☐ a) Compare nested models
- ☐ b) Detect outliers
- ☐ c) Validate stationarity
- ☐ d) Measure multicollinearity

✗ **Incorrect** 0/1 Points

37. A coefficient's 95% confidence interval spanning zero indicates:

- ☐ a) Statistical significance
- ☐ b) Practical significance
- ☐ c) Insufficient sample size
- ☐ d) Non-significant effect

✗ **Incorrect** 0/1 Points

38. The primary purpose of a "holdout sample" is to:

- ☐ a) Reduce overfitting
- ☐ b) Test model generalizability
- ☒ c) Handle missing data
- ☐ d) Normalize variables

✗ **Incorrect** 0/1 Points

39. **Which MMM step is most prone to subjective bias?**

- ☐ a) Data collection
- ☐ b) Variable selection
- ☐ c) Model validation
- ☐ d) ROI calculation

✗ **Incorrect** 0/1 Points

40. **The "Eigenvalue" in factor analysis measures:**

- ☐ a) Variance explained by a factor
- ☐ b) Multicollinearity severity
- ☐ c) Residual autocorrelation
- ☐ d) Model fit

✓ **Correct** 1/1 Points

41. **A media channel with high ROI but low elasticity should:**

- ☐ a) Receive increased budget
- ☒ b) Be capped at current spend
- ☐ c) Be tested for interaction effects
- ☐ d) Be excluded from the model

✗ **Incorrect** 0/1 Points

42. **The "Rubin Causal Model" in MMM is used to:**

- ☐ a) Estimate counterfactuals
- ☐ b) Reduce multicollinearity
- ☐ c) Validate adstock rates
- ☐ d) Optimize budgets

✗ **Incorrect** 0/1 Points

43. **AIC (Akaike Information Criterion) prioritizes models with:**

- ☐ a) Fewer parameters and better fit

- ☐ b) More parameters and better fit
- ☐ c) Fewer parameters regardless of fit
- ☐ d) Lowest p-values

✗ **Incorrect** 0/1 Points

44. **Which MMM challenge is exacerbated by cookie deprecation?**

- ☐ a) Attribution accuracy
- ☐ b) Data granularity
- ☐ c) Model interpretability
- ☐ d) Seasonality adjustment

✗ **Incorrect** 0/1 Points

45. **The "Janus coefficient" validates:**

- ☐ a) Model stability over time
- ☐ b) Adstock decay rates
- ☐ c) ROI accuracy
- ☐ d) Multicollinearity

✓ **Correct** 1/1 Points

46. **A media mix with a Herfindahl-Hirschman Index (HHI) of 0.7 suggests:**

- ☐ a) Perfect competition
- ☒ b) High concentration
- ☐ c) Moderate diversification
- ☐ d) Low risk

✗ **Incorrect** 0/1 Points

47. **Which optimization method is best for non-linear MMM problems?**

- ☐ a) Linear programming
- ☒ b) Gradient descent
- ☐ c) Simulated annealing
- ☐ d) Integer programming

✓ **Correct** 1/1 Points

48. **The "degrees of freedom" in a regression model represent:**

- ☒ a) Number of observations minus predictors
- ☐ b) Model flexibility
- ☐ c) Variance explained

☐ d) Residual sum of squares

✓ **Correct** 1/1 Points

49. . A media plan's "effective frequency" refers to:

☒ a) Minimum exposures needed for impact

☐ b) Optimal ad duration

☐ c) Click-through rate

☐ d) Cost per acquisition

✓ **Correct** 1/1 Points

50. Which metric is least relevant for MMM validation?

☐ a) R-squared

☐ b) RMSE (Root Mean Squared Error)

☒ c) AUC-ROC

☐ d) MAE (Mean Absolute Error)

✗ **Incorrect** 0/1 Points

51. A difference-in-differences (DiD) approach in MMM is used to:

- ☐ a) Isolate the impact of a marketing intervention
- ☐ b) Reduce multicollinearity
- ☐ c) Forecast long-term trends
- ☐ d) Optimize budgets

✗ **Incorrect** 0/1 Points

52. **Which term describes the use of MMM outputs to guide marketing experiments?**

- ☐ a) Closed-loop optimization
- ☒ b) Media flighting
- ☐ c) Conjoint analysis
- ☐ d) Sentiment analysis

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